

Arjuna JEE 2025

Mathematics

Straight Lines

DPP: 5

- Q1** Find the distance between the parallel lines $3x - 4y + 7 = 0$ and $3x - 4y + 5 = 0$.
- Q2** The distance between the parallel lines $y = 2x + 4$ and $6x = 3y + 5$ is -
 (A) $17/\sqrt{3}$
 (B) 1
 (C) $3/\sqrt{5}$
 (D) $17\sqrt{5}/15$
- Q3** Orthocentre of the triangle whose sides are given by $4x - 7y + 10 = 0$, $x + y - 5 = 0$ and $7x + 4y - 15 = 0$
 (A) $(-1, -2)$
 (B) $(1, -2)$
 (C) $(-1, 2)$
 (D) $(1, 2)$
- Q4** A line has slope m and y -intercept 4. The distance between the origin and the line is equal to
 (A) $\frac{4}{\sqrt{1-m^2}}$
 (B) $\frac{4}{\sqrt{m^2-1}}$
 (C) $\frac{4}{\sqrt{m^2+1}}$
 (D) $\frac{4m}{\sqrt{1+m^2}}$
- Q5** If the equation of base of an equilateral triangle is $2x - y = 1$ and the vertex is $(-1, 2)$, then the length of the sides of the triangle is
 (A) $\sqrt{20/3}$
 (B) $2/\sqrt{15}$
 (C) $\sqrt{8/15}$
 (D) $\sqrt{15/2}$
- Q6** The lengths of the perpendiculars from the points $(m^2, 2m)$, $(mn, m + n)$ and $(n^2, 2n)$ to the line $x + \sqrt{3}y + 3 = 0$ are in
 (A) Arithmetic progression
 (B) Geometric progression
 (C) Harmonic progression
 (D) None of these
- Q7** The equations of two sides of a square whose area is 25 square units are $3x - 4y = 0$ and $4x + 3y = 0$. The equations of the other two sides of the square are
 (A) $3x - 4y \pm 25 = 0$, $4x + 3y \pm 25 = 0$
 (B) $3x - 4y \pm 5 = 0$, $4x + 3y \pm 5 = 0$
 (C) $3x - 4y \pm 5 = 0$, $4x + 3y \pm 25 = 0$
 (D) None of these
- Q8** Lines are drawn parallel to the line $4x - 3y + 2 = 0$, at a distance $\frac{3}{5}$ from the origin.
 Then which one of the following points lies on any of these lines?
 (A) $(\frac{1}{4}, -\frac{1}{3})$
 (B) $(-\frac{1}{4}, \frac{2}{3})$
 (C) $(-\frac{1}{4}, -\frac{2}{3})$
 (D) $(\frac{1}{4}, \frac{1}{3})$
- Q9** The line L given by $\frac{x}{5} + \frac{y}{b} = 1$ passes through the point $(13, 32)$. The line K is parallel to L and has the equation $\frac{x}{c} + \frac{y}{3} = 1$. Then the distance between L and K is



- (A) $\sqrt{17}$
 (B) $\frac{17}{\sqrt{15}}$
 (C) $\frac{23}{\sqrt{17}}$
 (D) $\frac{23}{\sqrt{15}}$

Q10 One diagonal of a square is $3x - 4y + 8 = 0$ and one vertex is $(-1, 1)$, then the area of square is

- (A) $1/50$ sq. unit
 (B) $1/25$ sq. unit
 (C) $3/50$ sq. unit
 (D) $2/25$ sq. unit

Q11 The equation of base of an equilateral triangle is $x + y = 2$. The vertex is $(2, -1)$ then area of triangle is-

- (A) $2\sqrt{3}$
 (B) $\frac{\sqrt{3}}{6}$
 (C) $\frac{1}{\sqrt{3}}$
 (D) $\frac{2}{\sqrt{3}}$

Q12 The points on the line $x + y = 4$ which lie at a unit distance from the line $4x + 3y = 10$, are

- (A) $(3, 1), (-7, 11)$
 (B) $(-3, 1), (-7, 11)$
 (C) $(3, 1), (7, 11)$
 (D) $(1, 3), (-7, 11)$

Q13 If the foot of the perpendicular from the origin to a straight line is at the point $(3, -4)$ then the equation of the line is

- (A) $3x - 4y = 25$
 (B) $3x - 4y + 25 = 0$
 (C) $4x + 3y - 25 = 0$
 (D) $4x - 3y + 25 = 0$

Q14 A line passes through $(2, 2)$ and is perpendicular to the line $3x + y = 3$. Its y -intercept is:

- (A) $1/3$ (B) $2/3$
 (C) 1 (D) $4/3$

Q15 If the lines $x + 2y + 3 = 0$, $x + 2y - 7 = 0$ and $2x - y - 4 = 0$ form three sides of a square, the equation of fourth side may be

- (A) $2x - y - 6 = 0$
 (B) $2x - y + 6 = 0$
 (C) $2x - y - 14 = 0$
 (D) $2x - y + 14 = 0$

Q16 The equation of a line parallel to $x + 2y = 1$ and passing through the point of intersection of the lines $x - y = 4$ and $3x + y = 7$ is

- (A) $x + 2y = 5$
 (B) $4x + 8y - 1 = 0$
 (C) $4x + 8y + 1 = 0$
 (D) None of these

Q17 If a, b, c are in A.P., then $ax + by + c = 0$ will always pass through a fixed point whose coordinates are-

- (A) $(1, -2)$ (B) $(-1, 2)$
 (C) $(1, 2)$ (D) $(-1, -2)$

Q18 The straight lines $ax + by + c = 0$ where $3a + 2b + 4c = 0$ are concurrent at the point

- (A) $(1/2, 3/4)$
 (B) $(3/4, 1/2)$
 (C) $(-3/4, -1/2)$
 (D) $(-3/4, 1/2)$

Q19 Find the fix point through which the line $x(a + 2b) + y(a + 3b) = a + b$ always passes for all values of a and b -

- (A) $(2, 1)$ (B) $(1, 2)$
 (C) $(2, -1)$ (D) $(1, -2)$

Q20 The equation of straight line through the intersection of the lines $x - 2y = 1$ and $x + 3y = 2$ and parallel to $3x + 4y = 0$, is

- (A) $x + 4y + 5 = 0$
 (B) $3x + 4y - 10 = 0$
 (C) $3x + 4y - 5 = 0$



(D) $3x + 4y + 6 = 0$

Q21 Given a family of lines

$(2x + y + 4) + \lambda(x - 2y - 3) = 0$, the number of lines belonging to the family at a distance $\sqrt{10}$ from $P(2, -3)$ is

- (A) 0 (B) 1
(C) 2 (D) 4

Q22 The fix point through which the lines

$(2 \cos \theta + 3 \sin \theta)x + (3 \cos \theta - 5 \sin \theta)y - (5 \cos \theta - 2 \sin \theta) = 0$ passes for all values of θ , is

- (A) (0, 0) (B) (1, 1)
(C) (2, 1) (D) None of these

Q23 If $16a^2 - 40ab + 25b^2 - c^2 = 0$, then the line $ax + by + c = 0$ passes through the points-

- (A) (4, -5) and (-4, 5)
(B) (5, -4) and (-5, 4)
(C) (1, -1) and (-1, 1)
(D) None of these

Q24 If the straight lines

$x + y - 2 = 0$, $2x - y + 1 = 0$ and $ax + by - c = 0$ are concurrent, then the family of

lines $2ax + 3by + c = 0$ (a, b, c are non-zero) is concurrent at

- (A) (2, 3)
(B) $\left(\frac{1}{2}, \frac{1}{3}\right)$
(C) $\left(\frac{-1}{6}, \frac{-5}{9}\right)$
(D) $\left(\frac{2}{3}, -\frac{7}{5}\right)$



Answer Key

Q1 2/5

Q2 (D)

Q3 (D)

Q4 (C)

Q5 (A)

Q6 (B)

Q7 (A)

Q8 (B)

Q9 (C)

Q10 (D)

Q11 (B)

Q12 (A)

Q13 (A)

Q14 (D)

Q15 (B, C)

Q16 (B)

Q17 (A)

Q18 (B)

Q19 (C)

Q20 (C)

Q21 (B)

Q22 (B)

Q23 (A)

Q24 (C)



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